

# PAM Detector Module 041

# Test Report and Datasheet

Built and tested March 2026 by Jay Frothingham and Colin Rush.

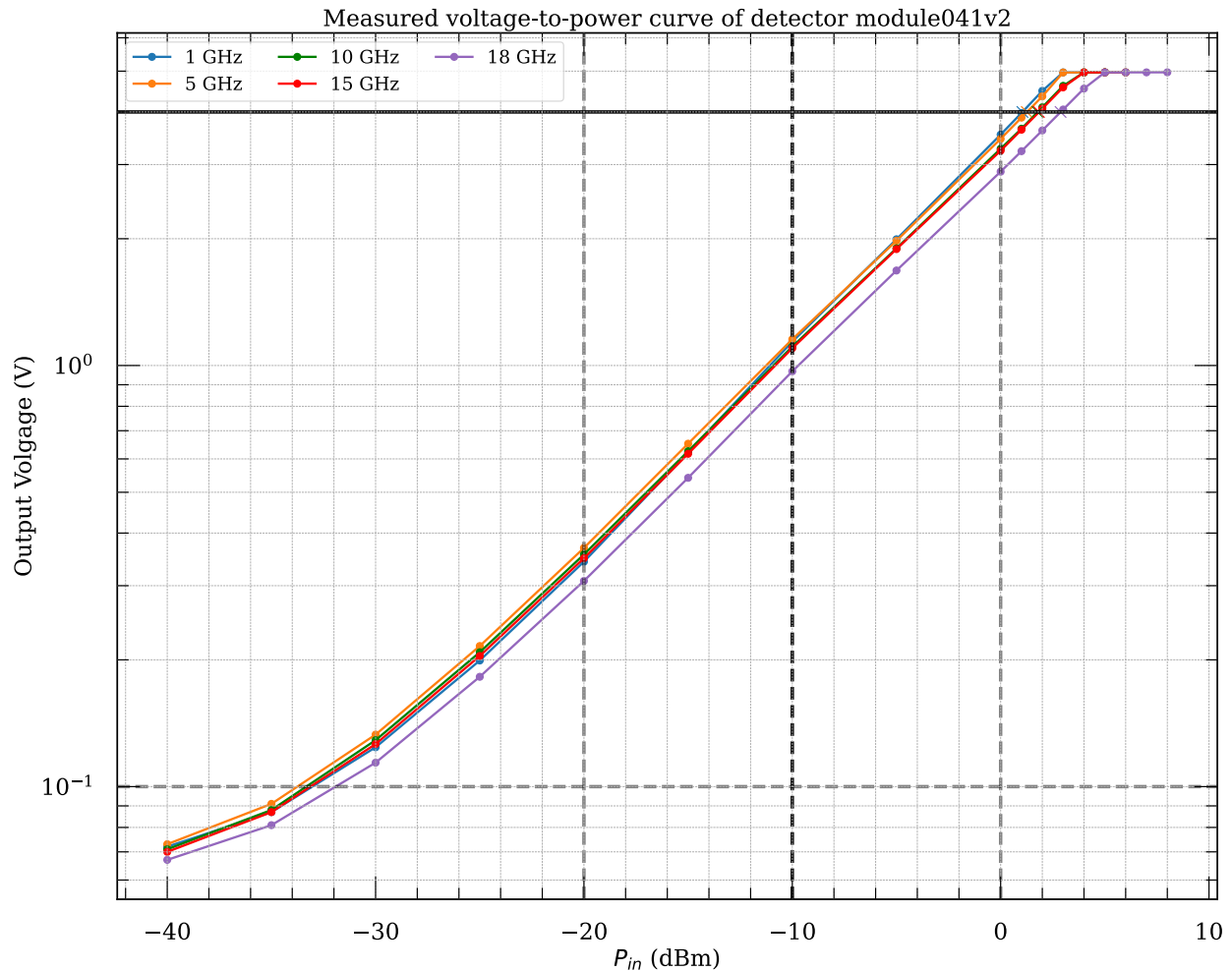
## Notes

First build failed due to impedance mismatch at low frequencies. Success after rework.

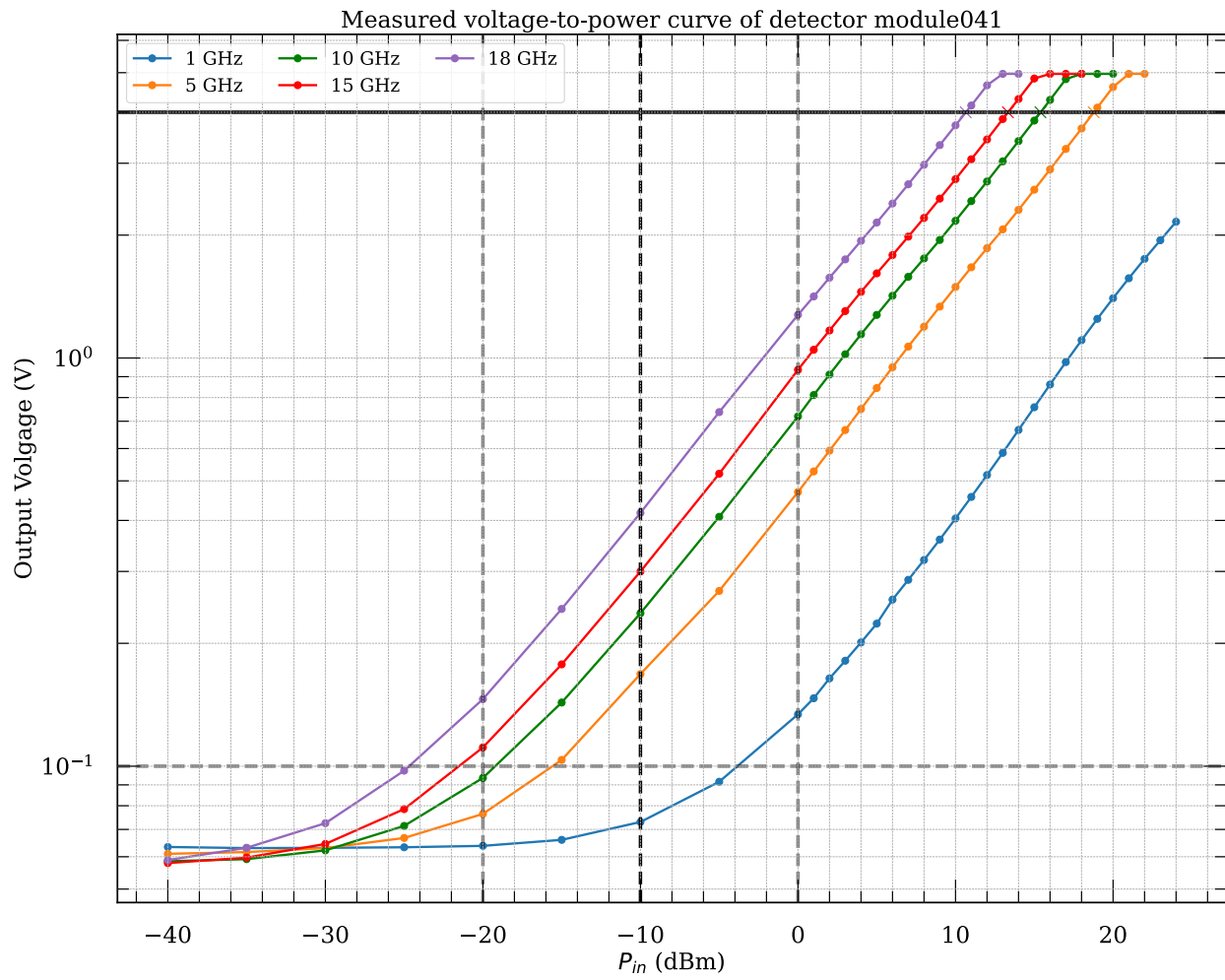
## Voltage-to-power Curve

A signal generator (E4420B) was used to apply an RF input with a fixed frequency and varying power levels. It was connected with a 14 inch long SMA cable (AFX-CA-141-14 AtlanTec RF) to the detector module. The power was increased stepwise and the corresponding output voltages were measured using a multimeter. A power supply provided the +5 V supply voltage for the detector module.

Reworked module P-V curves



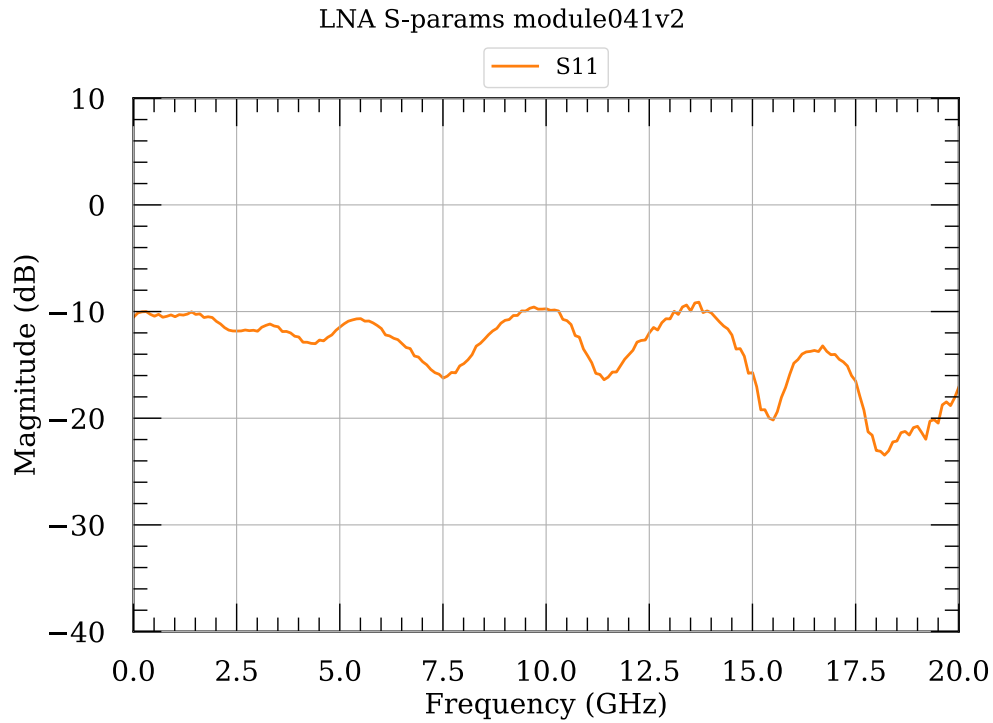
First build P-V curves.



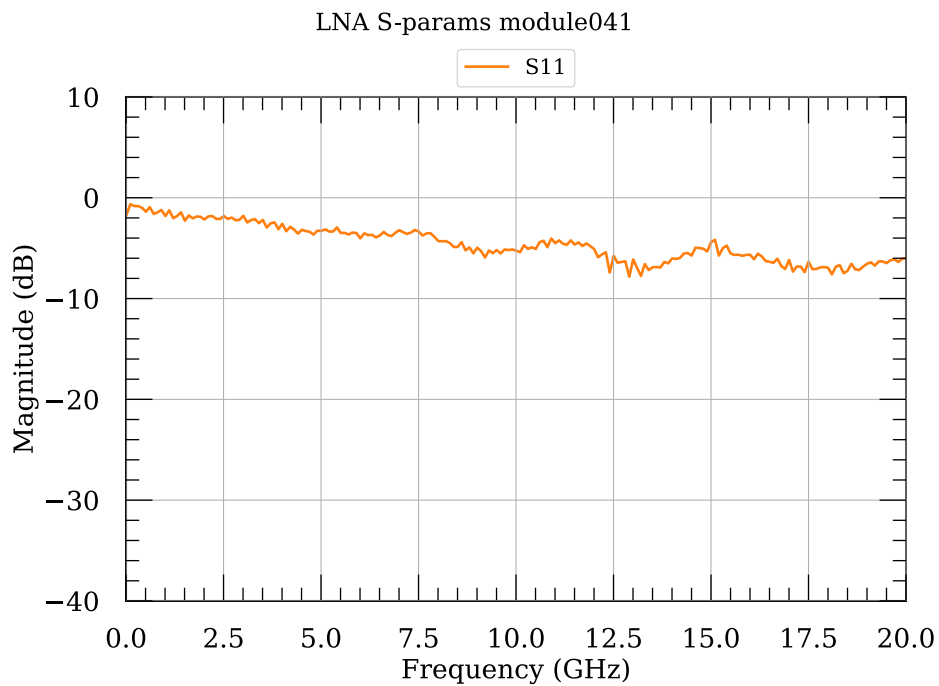
## S-parameter

S-parameter measurement with the VNA (N5230C).

Reworked module S-parameters



First build S-parameters

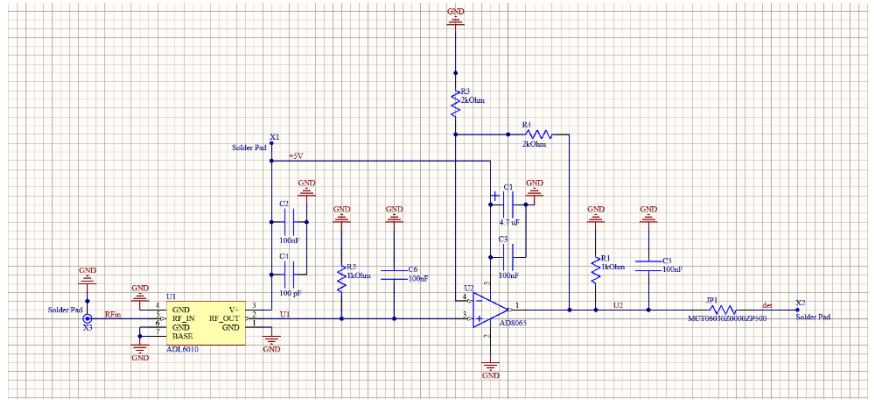


## Schematic

- Frequency: 0.5-20GHz
- Impedance: 50Ω

Inputs:

- RFin: Radio signal input
- +5V: positive supply voltage (+5V@1.6mA)
- GND



Outputs:

- Det: Output for power measurement

### Components, Footprint - All Case Codes are metric

- |                                            |             |
|--------------------------------------------|-------------|
| 1 – SMA 2Hole (142-1701-201)               | [X3]        |
| 2 – Feed through cap: 4-40UNC-2A (B3C153B) | [X1,X2]     |
| 1 – Turret Terminal 2-56 UNC – 2A (1595-2) |             |
| 1 – detector LFCSP-6 (ADL6010ACPZN-R2)     | [U1]        |
| 1 – op amp SOT-23-5 (AD8065ARTZ-REEL7)     | [U2]        |
| 1 – 100pF 1005 (GRM1555C2A101FA01D)        | [C4]        |
| 1 – 4.7uF 3216 (TH3A475K020C5000)          | [C1]        |
| 4 – 100nF 1608 (GCM188R71C104KA37J)        | [C2-3,C5-6] |
| 1 – 0Ω 1608 (MCT06030Z0000ZP500)           | [JP1]       |
| 1 – 1MΩ 1608 (MCT06030C1004FPW)            | [R1]        |
| 1 – 2kΩ 1608 (MCT06030C2001FP500)          | [R2]        |
| 1 – 1.6kΩ 1608 (MCT06030C1601FP500)        | [R4]        |
| 1 – 402Ω 1608 (MCT06030C4020FP500)         | [R3]        |

- 1 – RO4350 PCB  
1 – Box  
1 – Lid  
2 – Screws 3-48 UNC - 2B x 3/16 (92196A091)  
4 – Screws 2-56 UNC - 2B x 1/8 (21202)  
4 – Screws 2-56 UNC - 2B x 5/32 (91771A884)  
1 – RF-absorber PSA 0.08", ca. 20 x 24 mm (MR42-0008-20)

